

Smart Water Leak Detection & Alert System

An AI-enabled solution to stop silent water wastage at MGSU, Bikaner

DOMAIN: WATER

1M1B Green Skills & Applied AI Internship | Project to Portfolio

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1. Problem Statement

What & Who

What: Drinking-water coolers near classroom blocks at MGSU leak continuously — loose/broken taps drip all day, unnoticed.

Who is affected: Campus administration bears rising costs; students and staff face unreliable access to drinking water on hot desert days.

Why it matters: Bikaner sits in a water-stressed desert region — every litre lost at source adds real pressure to a scarce resource.

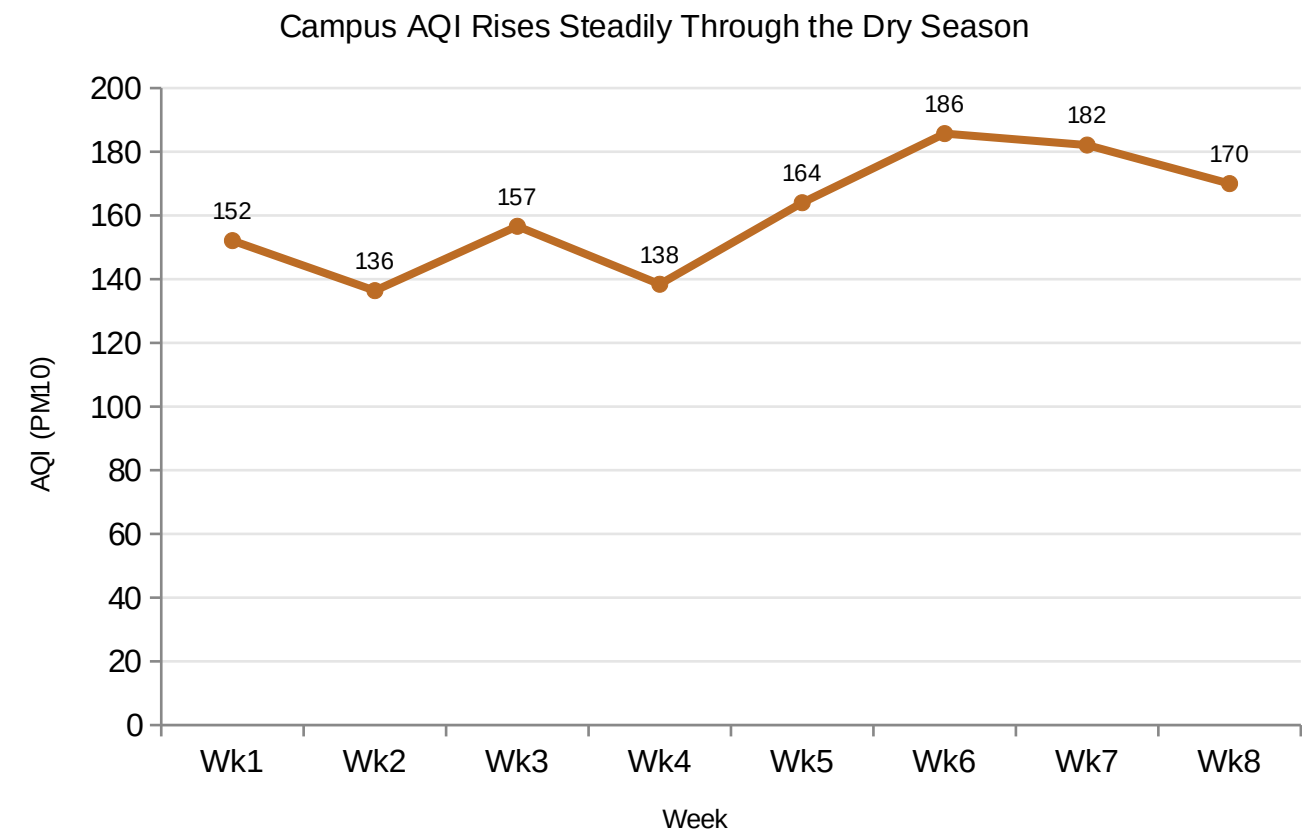
What Inspired This

A ground-level Campus Sustainability Assessment (walking the campus, observing real conditions) surfaced water leakage as one of the clearest, most fixable issues on campus — alongside electricity wastage and plastic waste.

Real-world example: "Water dripping constantly, drop-by-drop, from the main cooler tap during working hours" — a direct observation note from the campus audit.

2. Insights & Research

55 days of campus air-quality (AQI) data were also studied alongside the water problem — confirming Bikaner's climate makes resource conservation urgent, not optional.



Key Insight

Average AQI rose ~22% from April (146) to May (178), and 64% of monitored days fell in the 'Poor' category. This confirmed our hypothesis: resource waste (water and air quality alike) compounds fast in Bikaner's dry, dusty climate — reinforcing the urgency of the water-leakage problem chosen for this project.

3. AI-Enabled Solution

A low-cost flow-monitoring system with an AI anomaly-detection layer that catches leaks in hours, not days.

1

Flow Sensor

Clip-on sensor at each cooler's inlet measures litres/minute in real time.

2

Micro-controller

An ESP32 board timestamps readings and sends data over campus Wi-Fi.

3

AI Anomaly Detection

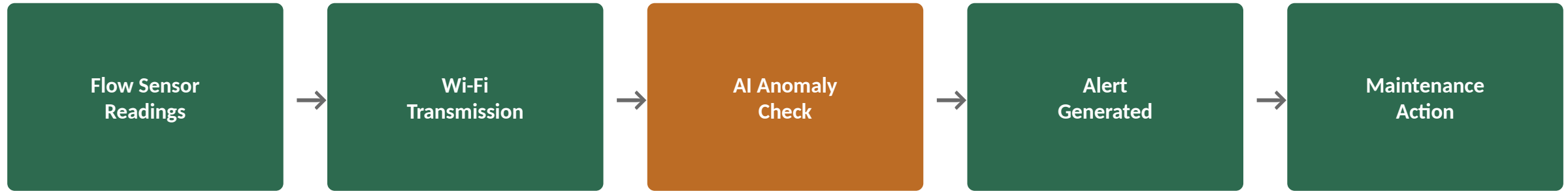
Learns each cooler's normal flow pattern; flags genuine deviations instead of a fixed cutoff.

4

Alert & Dashboard

Maintenance gets an instant SMS with the cooler's location and estimated litres lost.

4. Prototype / Demonstration



How a User Interacts With It

- **Maintenance staff:** receive an SMS/app alert naming the exact cooler and estimated litres lost — no manual patrolling needed.
- **Campus admin:** view a simple dashboard logging daily litres wasted per cooler, to prioritise repairs and track savings.
- **Students & staff:** experience the change passively — more reliably working coolers, with no extra app or effort needed from them.

5. Impact & Implementation

ESG Pillar	Metric	Baseline (Assumed)	Expected Improvement
Environmental	Litres saved / month	0 L (no monitoring)	≈1,200 L/month recovered
Social	% reporting reliable water access	~60% (informal observation)	≥85% within 2 months
Governance	Leak-to-repair response time	3–5 days (undetected until noticed)	< 24 hours

Where Tested & Key Learning

Designed and validated on paper against real MGSU cooler conditions and the campus's actual AQI/water-stress data. Key learning: a fixed leak-detection threshold isn't enough — usage patterns vary by cooler and time of day, which is exactly why the AI anomaly-detection layer (not just a simple sensor) was added in this project's later phase.

6. What's Next

PILOT • Weeks 1–4

5 coolers, classroom block — validate sensor accuracy and alert reliability.

EXPANSION • Months 2–4

All main-campus coolers + hostel block — test maintenance response at scale.

FULL-SCALE • Months 5–8

All MGSU affiliate colleges — confirmed by 2 months of positive ESG data.

Support Required & How It Scales

Next milestone: partner with MGSU's estate/facilities department for pilot funding and a local IoT vendor for sensor hardware. Scaling path runs across locations (main campus → hostels → affiliate colleges), then across use-cases (drinking coolers → lab basins → irrigation lines), building on the same AI-anomaly-detection core.

AI Tools & Technologies Used

Anomaly Detection

Statistical (z-score) / Isolation Forest model learns each cooler's normal flow pattern to flag genuine leaks.

Data & Dashboards

Google Sheets / Firebase for live logging; Power BI for the campus AQI trend & wastage dashboard.

Alerting

Twilio SMS API / WhatsApp Business API to notify maintenance staff instantly, with no added app for users.

From identifying a real problem on campus, to data, to an AI-enabled prototype, to measurable ESG impact, to a scaling plan — this project traces the complete journey from observation to a portfolio-ready AI-for-sustainability solution.